

Tech Exercise Preparation

[PCDex – PokeDex for PC parts]

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1. Description of the Web Application

For my project, I will develop a PC build and part tracker web application. This application will let the user log individual computer components, organize them into their own custom build, and quickly find parts by category or build.

The main functionality of the app includes adding, viewing, editing, and deleting parts. It will also be able to organize parts in to named builds and search for parts across all of them. Each part will have:

- Part Name: name or model of the component (Ryzen 7 7600X3D)
- Category: type of component (CPU, GPU, RAM, Motherboard, etc.)
- Brand: name of brand that made the component
- Price: Price of the component
- Build: the name of the build that you put this part into
- Notes: additional notes you want to add (clock speed, storage amount, refresh rate, etc.)

This PCDex will allow users to:

- View all the parts in an organized build
- Add new parts to their build
- Switch out parts from their build
- Delete parts from their build
- Search or filter parts by categories above

2. Links to Relevant Tutorials

In order to help me get started with this web application, I added helpful tutorials to help me understand the basics of Django, SQLite, and GitHub

Tutorial 1: Django Tutorial for Beginners – Build Powerful Backends (Programming with Mosh)

Link: <https://www.youtube.com/watch?v=rHux0gMZ3Eg>

This is a deep dive video that goes over how Django works and even goes through a starter project so I can have hands on learning with Django

Tutorial 2: SQLite Tutorial – Playlist by Coding Under Pressure

Link: https://youtube.com/playlist?list=PLJJcOjd3n1ZfwZKYM_uJSs0xSejPsQlav&si=jyGF_Wx_LBPQXPEOX

This is a playlist that goes over many different ways to create, use, and integrate SQLite which I will pick and choose which video I would need depending on what my app needs

Tutorial 3: GitHub Docs – Getting Started with GitHub Documentation

Link: <https://docs.github.com/en/get-started>

This official guide from GitHub explains how to create repositories, commit changes, and push code. GitHub Docs is also a great website that I can search different questions and it will provide me with documentation on how to solve it.

3. Web Framework, Database, and Version Control System

For my web application, I plan to use:

Web Framework: Django – Django is a high-level Python web framework that handles routing, form processing, and database access through its ORM. It will manage the “Add Part” form, the parts list view, and the search/filter

Database: SQLite – SQLite is a lightweight, file-based database that requires no separate server setup and comes built into Django which makes it very convenient and practical

Version Control: GitHub – I will put my source code in a GitHub repository, committing changes regularly so I can go through the project iteratively and if I mess up I have backups of previous working versions.

Django’s ORM will handle storing new parts, updating existing ones, and querying the database when a user searches by category, brand, or build name, so I will not need to write raw SQL for most operations.

4. Cloud Hosting Service

For hosting the web application, I will use:

Cloud Hosting: Amazon AWS – AWS will provide a virtual server where I can install and run my Django application publicly so others can use it outside of my local computer

Operating System: Ubuntu – Ubuntu is a stable, widely documented Linux distribution that is well suited for running Python web applications on AWS

I will start by developing on my local device using Django’s built in development server, and once the application works then I will deploy it to an Ubuntu based AWS instance.

5. Development Environment

For development I will use:

IDE: Visual Studio Code – VSCode is a lightweight editor with many Python and Django implementations, and extensions

Python & Django: I will use Python 3 and Django in a venv (virtual environment) and use Django’s built in development server for local testing before deploying on AWS